GIT Installation

- Before using GIT commands, GIT must be installed on the computer. After installation, the GIT command can be used from a terminal or command prompt.

What is GIT Installation

GIT Installation is the process of installing the GIT software on a computer so that GIT commands can be executed and repositories can be managed.

Installing GIT on Windows

The basic installation steps are:

1. Open a web browser.
2. Go to the official GIT website.
3. Download the GIT installer for Windows.
4. Run the downloaded installer.
5. Follow the installation wizard.
6. Keep the default options unless a specific configuration is required.
7. Complete the installation.

Verify GIT Installation

After installation, open Command Prompt, PowerShell, Git Bash, or another terminal and execute:

```
git --version
```

If GIT is installed correctly, the command displays the installed GIT version.

```
git version 2.x.x
```

The exact version number depends on the version installed on the computer.

GitHub Account Creation

- GitHub is a platform used to host GIT repositories and support software-development collaboration. A GitHub account allows a user to create and manage repositories on GitHub.

GIT and GitHub

GIT	GitHub
Version control software	Online platform for hosting and collaborating GIT repositories.

Runs on the developer's computer.	Provides repositories and collaboration features through an online service.
Tracks changes and creates commits.	Can store and share repositories remotely.

Steps to Create a GitHub Account

1. Open a web browser.
2. Go to the official GitHub website.
3. Select the option to create a new account.
4. Enter the required email address.
5. Create a password.
6. Choose a GitHub username.
7. Complete the required verification steps.
8. Verify the email address if requested.
9. Complete the account setup.

Choosing a GitHub Username

A username identifies the user's GitHub account and appears in the user's GitHub profile and repository URLs.

- Choose a professional and easy-to-remember username.
- Avoid unnecessarily complicated names.
- Remember that the username can become part of repository links.

GitHub Repository

A repository on GitHub is an online location where a project's files and GIT history can be stored.

For example, a student may create a repository named:

```
student-management-project
```

GIT Configuration

- After installing GIT, basic configuration should be performed before creating commits. The most important configuration is the user's name and email address.

Why GIT Configuration is Needed

- GIT records the author information along with commits. Configuring the user name and email helps identify who created a particular commit.

Check Existing Configuration

To display the current GIT configuration, use:

```
git config --list
```

To check the configured user name:

```
git config user.name
```

To check the configured email:

```
git config user.email
```

Configure User Name

Use the following command:

```
git config --global user.name "Your Name"
```

Configure Email

Use the following command:

```
git config --global user.email "you@example.com"
```

The --global option applies the configuration to the user's GIT configuration for repositories on that computer.

Example

```
git config --global user.name "abc"
git config --global user.email "abc@gmail.com"
```

Verify Configuration

After configuring the name and email, verify them using:

```
git config --global user.name
git config --global user.email
```

Local Repository Configuration

GIT also allows configuration to be applied only to a particular repository. This is useful when a repository needs settings different from the global configuration.

```
git config user.name "Project User"
```

```
git config user.email "project@example.com"
```

A repository-specific configuration takes precedence over the corresponding global setting for that repository.

Global vs Local Configuration

Configuration	Scope	Command Example
Global	Applies to repositories for the current user on the computer.	git config --global user.name "Your Name"
Local	Applies only to the current repository.	git config user.name "Project User"

Useful GIT Configuration Commands

Command	Purpose
git config --list	Displays configuration settings.
git config user.name	Displays the name configured for the current repository.
git config user.email	Displays the email configured for the current repository.
git config --global user.name "Name"	Sets the global user name.
git config --global user.email "Email"	Sets the global email.